

Evaluation metric

R2 score:

$$1 - \frac{SS_{res}}{SS_{total}}$$

SS_{total} $\rightarrow$ sum of squared of error of a mean predictor

$$= \sum_{i=1}^n (y_i - \bar{y})^2$$

SS_{res} $\rightarrow$ sum of squares of error of a learnt predictor

$$= \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

when R^2 score = 0, learnt model is equivalent to the base model

when $\frac{SS_{res}}{SS_{total}} \rightarrow 0$, means $SS_{res} \ll SS_{total}$

then R^2 score $\rightarrow 1$.

* Multivariate Linear regression:

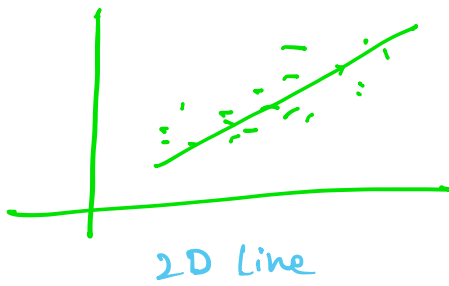
$$y = \beta_1 x + \beta_0 \rightarrow \text{Eqn for univariate.}$$

now we have input = $\{ (x_1^1, x_2^1, \dots, x_n^1) (x_1^2, x_2^2, \dots, x_n^2) \dots (x_1^m, x_2^m, \dots, x_n^m) \}$
Output = $\{ (y_1, y_2, \dots, y_m) \}$

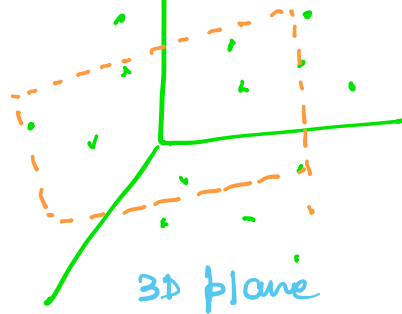
n dimensions, m samples.

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n$$

→ bias weights

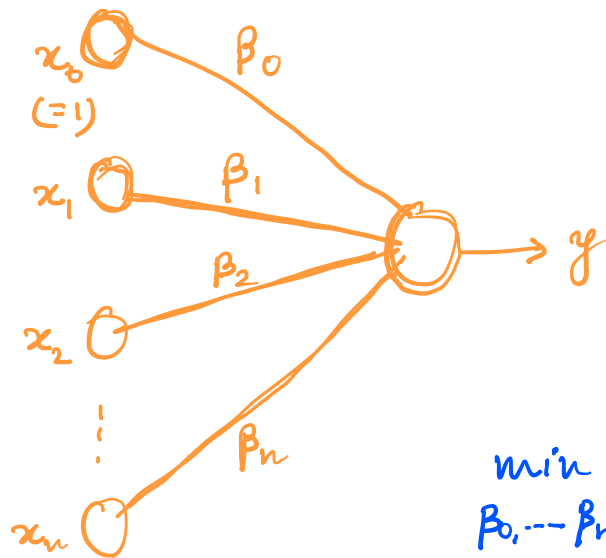


2D Line



3D plane

nD hyperplane



$$\hat{y} = 1 \cdot \beta_0 + x_1 \beta_1 + x_2 \beta_2 + \dots + x_n \beta_n$$

$$\text{Error} = \frac{1}{m} \sum_{i=1}^m (y_i - \hat{y}_i)^2$$

min $\beta_0, \dots, \beta_n$ (Error)

input matrix

$$X = \begin{bmatrix} x_1^1 & x_1^2 & x_1^3 & \dots & x_1^n \\ x_2^1 & x_2^2 & x_2^3 & \dots & x_2^n \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ x_m^1 & x_m^2 & x_m^3 & \dots & x_m^n \end{bmatrix}_{m \times n}$$

output matrix

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_m \end{bmatrix}_{m \times 1}$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_n \end{bmatrix}_{n \times 1} \rightarrow w = \begin{bmatrix} w_0 \\ w_1 \\ \vdots \\ w_n \end{bmatrix}_{n \times 1}$$

$$\hat{y} = xw$$

$m \times 1 \leftarrow m \times n \quad n \times 1$

$$e = (y - \hat{y})^2 \quad \Bigg| \quad \text{Error} = \underline{(Y - XW)^T (Y - XW)}$$

$$\begin{aligned} X^T \hat{y} &= X^T X W \\ X^T \hat{y} &= W \end{aligned} \quad \text{?}$$

$$X^T \hat{y} = \underline{X^T X} W$$

$$X \Rightarrow m \times n$$

$$X^T \Rightarrow n \times m$$

$$X^T X \Rightarrow n \times n$$

$$(X^T X)^T X^T \hat{y} = \underline{(X^T X)^T X^T X} W$$

$$(X^T X)^T X^T \hat{y} = W$$

$$J(W) = \frac{1}{n} (Y - XW)^T (Y - XW)$$

Error / loss function.

Gradient

$$\frac{\partial J}{\partial W} = -\frac{2}{n} X^T (Y - XW)$$

weight
update

$$W_{\text{new}} = W_{\text{old}} - \eta \cdot \frac{\partial J}{\partial W}$$

Assumptions:

- ① X and y relationship is linear in nature
- ② Data distribution is IID.
- ③



Cat $\rightarrow$ a, b, c

	Cat
1	a
2	b
3	c

cat_a	cat_b	cat_c
1	0	0
0	1	0
0	0	1

